Unusual Manifestations of the Trigeminocardiac Reflex During Awake Craniotomy: A Case Report and Review of the Literature

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The trigeminocardiac reflex (TCR) is triggered by stimulation of a branch of the trigeminal nerve and results in vagally mediated bradycardia, hypotension, apnea, and gastrointestinal hypermotility. In the operating theatre, patients susceptible to TCR are typically under general anesthesia; thus, cardiac abnormalities are the most common manifestation. Our case highlights the less common intraoperative manifestations of gastric hypermotility and apnea in a patient undergoing awake craniotomy for tumor resection. Prompt recognition, removal of stimuli, and airway management prevented catastrophic complications while facilitating completion of the procedure. (A&A Practice. 2021;15:e01508.)

GLOSSARY

BP = blood pressure; **BPM** = beats per minute; **HIPAA** = Health Insurance Portability and Accountability Act; **HR** = heart rate; **LMA** = laryngeal mask airway; **MAP** = mean arterial pressure; **RR** = respiratory rate; **TCR** = trigeminocardiac reflex

The trigeminocardiac reflex (TCR) occurs when stimulation of a trigeminal nerve branch causes a sudden and profound parasympathetic response. The oculocardiac reflex subtype of TCR was first described in 1908 and has been recognized as a common occurrence during procedures provoking the ophthalmic branch of the trigeminal nerve, resulting in bradycardia or even asystole.[1] Schaller et al[2] first described central trigeminal nerve stimulation during neurosurgery as a potent trigger of TCR. In the 2 decades since, there has been a growing interest in the pathophysiology, classification and presentation of TCR.

The afferent limb of TCR can be activated by insults such as stretch, pressure, temperature, and electrical stimulation.[3] Impulses flow from a sensory branch through the Gasserian ganglion to the sensory nuclei of the trigeminal nerve.[3,4] Small internuncial fibers of the reticular formation facilitate the brainstem reflex to activate the efferent limb, which is primarily mediated by the dorsal motor nucleus of the vagus and results in parasympathetic effects including bradycardia, hypotension, apnea, and gastric hypermotility (Figure).[3,4] Interestingly, reflex sympathetic activity has also been described with peripheral trigeminal stimulation, causing both parasympathetic and sympathetic responses that manifest with normotension or hypertension.[3,5,6] The diving reflex, another subtype of TCR characterized by bradycardia, apnea, and hypertension on immersion of the head in cold water, exemplifies this mixed response.[3]

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Current definitions of TCR require a 20% decrease in heart rate (HR); however, concomitant decreases in mean arterial pressure (MAP) remain flexible to include all subtypes of TCR.[5,7] Furthermore, TCR episodes should be plausible (ie, appearing promptly after trigeminal stimulation) and reversible (ie, abolished on stimulus removal).[6] Classification schemes group subtypes based on stimulation site and subsequent physiologic changes.[7] Consistent characterization has allowed clinicians to better understand, anticipate, and treat potentially life-threatening complications of TCR.

Numerous examples of intraoperative TCR have been reported during ophthalmologic and neurosurgical procedures.[1,7] Recently, unique cases of TCR have been described, including during reduction of a pan facial fracture, retraction of the temporalis muscle, and posttetanic-count stimulation of the facial nerve.[8–10] Nearly all episodes of TCR occurred while the patient was under general anesthesia and intubated. Moreover, symptoms were predominantly cardiac in nature. To our knowledge, only 2 prior accounts of TCR during awake craniotomy have been reported with primary manifestations of bradycardia progressing to asystole.[4,11] This case highlights an unusual presentation of TCR during awake craniotomy in which the patient experienced gastric hypermotility and apnea. The patient's family provided written Health Insurance Portability and Accountability Act (HIPAA) authorization to publish this case report.

CASE DESCRIPTION

A 75-year-old man with prior coronary artery bypass grafting presented with new-onset right homonymous hemianopsia, word-finding difficulty, and impaired memory. Imaging revealed a peripherally enhancing 8 × 5 × 5 cm mass centered on the left temporo-occipital junction abutting the occipital horn of the left lateral ventricle. Biopsy showed high-grade glioblastoma multiforme. Neuropsychological testing demonstrated semantic paraphasias, impaired picture naming, and impaired reading and writing. The decision was made to proceed with awake left temporal craniotomy for tumor removal.

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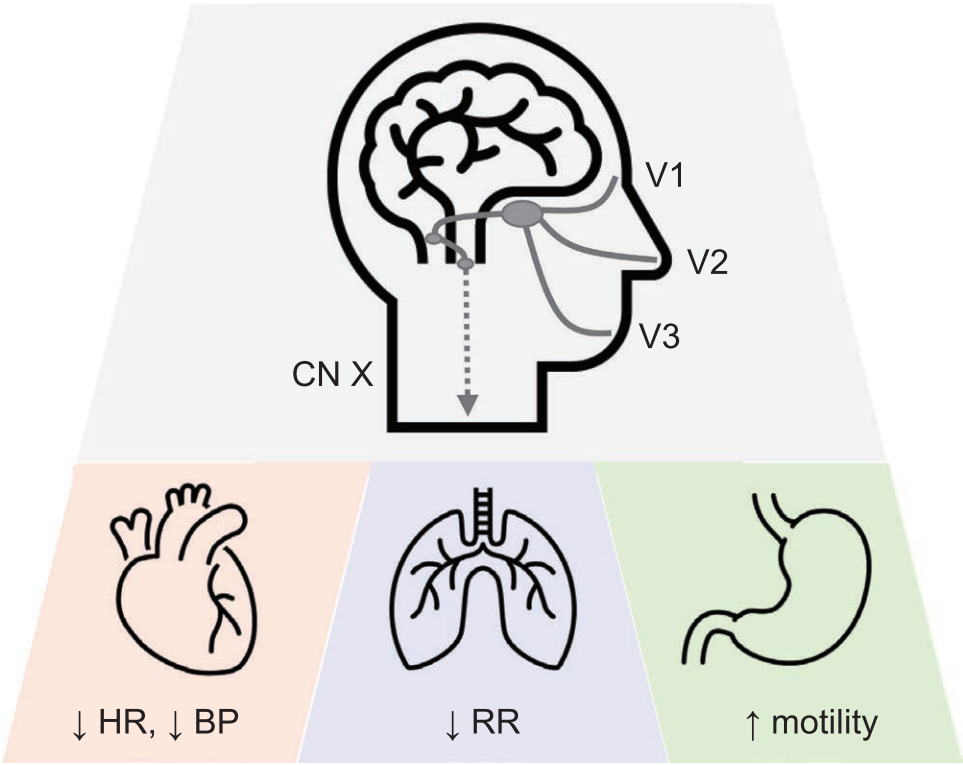


Figure. The trigeminocardiac reflex is activated by sensory stimulation of the trigeminal nerve (V1, V2, V3), with signals flowing through the Gasserian ganglion to the sensory nuclei of the trigeminal nerve. Efferent signals are mediated by the vagus nerve (CN X) and may result in parasympathetic end-organ effects such as bradycardia, hypotension, apnea, and gastric hypermotility. BP indicates blood pressure; HR, heart rate; RR, respiratory rate.

The patient was brought to the operating room, positioned supine, and standard monitors were placed. A left radial arterial line was established. $Etco_2$ was measured via nasal cannula. Baseline HR was 65 beats per minute (BPM) and blood pressure was 140/80 mm Hg with MAP of 100 mm Hg.

Infusions of propofol (10–20 µg/kg/min), dexmedetomidine (0.2–0.5 µg/kg/h), and remifentanil (0.01–0.03 µg/kg/min) were titrated to establish moderate sedation. Bilateral supratrochlear, supraorbital, zygomaticotemporal, auriculotemporal, lesser and greater occipital nerve blocks were performed using anatomic landmark guidance according to our modifications[12]; total infiltration volume was 30 mL local anesthesia mixture (25 mL 1% lidocaine with 1:200,000 epinephrine and 5 mL 1% tetracaine). The patient's head was fixed in place using a Mayfield holder with the neck partially flexed and rotated to the right. The patient was prepped and draped. An incision starting behind the left ear vertically and curving forward toward the anterior scalp was made. On exposure of the cortical surface, all sedation was stopped. Subsequent dissection and resection proceeded with direct electrical stimulation based on mapping using picture naming and reading paradigms.

One hour later, the patient endorsed abdominal discomfort with the urge to vomit, followed by small-volume emesis. The surgeon stopped momentarily and symptoms resolved. Thirty minutes later, a similar episode occurred with the patient retching but only spitting out saliva, again with resolution once the surgeon paused; 4 mg ondansetron was administered. Half an hour later, the patient felt unwell and unexpectedly became apneic without improvement in respiratory effort to loud voice or painful stimulus. Given the patient's head and neck position within the Mayfield apparatus, a laryngeal mask airway (LMA) was emergently placed to facilitate airway management. We ensured the LMA was well-seated with no leak at 20 cm H_2O. A spontaneous respiratory pattern with measurable $Etco_2$ quickly returned. Oxygen saturation remained 100% throughout the apneic episode.

After discussion with the surgeon, we determined that each episode coincided with surgical stimulation of one area of the dura and were likely manifestations of TCR. Review of the anesthetic record showed simultaneous drops in HR from approximately 70 to 50 BPM during all 3 events. Additionally, MAP decreased from approximately 90 to 60 mm Hg during the brief apneic episode. While additional stimulation at that site was not expected, neuropsychological testing was complete and the patient was tolerating the LMA, so we decided to keep it in place while restarting sedation.

The patient remained stable without further episodes of apnea, emesis, bradycardia, or hypotension through the remainder of the procedure. Electroencephalogram showed brief epileptic activity that was terminated with propofol bolus, otherwise the surgery proceeded uneventfully. The patient was extubated and monitored in the intensive care unit. His postoperative course was complicated by persistent aphasic seizures that were treated with lacosamide and levetiracetam. He was discharged home on postoperative day 4. During initial follow-up, he was recovering well with no obvious seizures. He planned to undergo chemo- and radiotherapy at a facility closer to home.

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DISCUSSION

This case features an uncommon presentation of TCR unique to patients undergoing awake procedures. Although apnea and gastric hypermotility are well-defined components of the vagomimetic reflex, most prior reports and current working models focus on cardiac aberrations. While our patient met the definition of TCR by HR and MAP criteria, the less prominent cardiac changes delayed diagnosis. Importantly, abdominal cramping, nausea, and vomiting may be early warning signs of more significant TCR events. Fortunately, TCR was eventually recognized and significant morbidity was avoided.

From a classification standpoint, this reflex arc represents the peripheral subtype; MAP remained stable for 2 of the 3 episodes, suggesting coactivation of the parasympathetic and sympathetic nervous systems. The afferent limb was likely activated by stimulation of a meningeal branch of the ophthalmic or mandibular division of the trigeminal nerve.

Risk factors for TCR include young age, increased resting vagal tone, male sex, and preoperative beta blocker and calcium channel blocker use.[3] Intraoperatively, hypercapnia, hypoxemia, and inadequate depth of anesthesia can precipitate TCR.[3] Anesthetic choice can also play a role. TCR is more often seen with potent opioids such as remifentanil, and when combined with other sympatholytic drugs like dexmedetomidine, the effect can be more pronounced.[4,13] Interestingly, TCR is less commonly seen with sevoflurane, ketamine, and propofol.[13] Clearly, the nature of an awake craniotomy as well as the anesthetic choice placed our patient at a high risk.

Although there are no definitive recommendations for prevention, if a patient is deemed high risk, premedication with an anticholinergic such as atropine or glycopyrrolate can be considered while taking into account medication side effects.[3] Branches of the trigeminal nerve near the surgical site can also be infiltrated with local anesthetic.[14] Additionally, the surgical approach should attempt to minimize direct trauma to the nervous tissue; when necessary, manipulation should be smooth and gentle.[3]

If TCR occurs, the most important step is to remove the stimulus. If bradycardia or hypotension persist, an anticholinergic drug should be administered while ruling out other causes of hemodynamic compromise. Epinephrine may be required to restore normal cardiac parameters.[15] In the case of an awake patient, endotracheal intubation should be considered to prevent apnea as well as protect the airway from potential aspiration following emesis. In this case, the patient was apneic and, given his head and neck position, emergent LMA placement allowed us to effectively manage the airway and avoid further morbidity. If additional surgical stimulation could not be avoided or the patient were actively vomiting, endotracheal tube placement would have been indicated. We were prepared to place an endotracheal tube via the LMA under fiberoptic guidance if needed.

In summary, this is the third reported case of TCR during awake craniotomy and the first case with primary manifestations of gastric hypermotility and apnea. An awake, unintubated patient poses unique challenges in the diagnosis and management of TCR. As anesthesiologists care for an increasing number of patients undergoing awake craniotomy, the prevalence of TCR in this vulnerable population will continue to rise. Appropriate anticipation, prompt recognition, effective communication, and rapid treatment may prevent major morbidity due to intraoperative TCR.

DISCLOSURES

Name: Anirban Sahu, MD, PhD.
Contribution: This author helped care for the patient and wrote the article.
Name: Antoun H. Koht, MD.
Contribution: This author helped care for the patient and wrote the article.
This manuscript was handled by: Mark C. Phillips, MD.

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